

WEEK 3 · DAY 1

VPC Fundamentals

BaseStack Academy · AWS Cloud Accelerator

Networking & VPC

Cohort 1

Beginner-Friendly

SAA-C03 Aligned

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Your VPC (10.0.0.0/16)



Internet Gateway



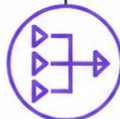
Public Subnet (10.0.1.0/24)

Web Servers - Bastion Host



Private Subnet (10.0.2.0/24)

App Servers - Databases



NAT Gateway

1 Overview — Why VPCs Are the Foundation of Everything

FOUNDATION

2 Key Concepts — VPC, CIDR, Subnets, Route Tables

CORE KNOWLEDGE

3 Key Concepts — IGW, NAT Gateway, Security Groups, NACLs

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EXAM PREP

Overview — Why VPCs Are the Foundation of Everything

When you launch an EC2 instance, create an RDS database, deploy a Lambda function with a private endpoint, or build any meaningful AWS application — every single one of those resources lives inside a network. That network is the Amazon Virtual Private Cloud, or VPC.

A VPC is your own logically isolated section of the AWS cloud. It looks, behaves, and is configured exactly like a traditional on-premise data centre network — but you manage it entirely through software, with no physical hardware to purchase, rack, or cable.

Without understanding VPCs, you cannot reason about network security, high availability, internet access, private connectivity, or cost architecture. Everything in Weeks 3 and 4 — EC2, RDS, Load Balancers, ECS, Lambda with private endpoints — sits on top of VPC foundations.

WHY NOW?

Every AWS Solutions Architect interview includes VPC questions. SAA-C03 Domain 1 (Design Resilient Architectures) and Domain 2 (Design Secure Architectures) both require deep VPC knowledge — specifically subnet design, routing, and security layer differences.

Nigerian banks and fintechs deploying on AWS must meet CBN requirements for network segmentation — production databases in private subnets, no direct internet exposure. Understanding VPC is the difference between compliant and non-compliant cloud architecture.

The difference between a junior cloud engineer and a mid-level one is largely VPC. Build the mental model now.

Beginner Note

Think of a VPC like the fence around your company's private estate in Lekki Phase 1. Inside the fence, you control what can move where. Outside the fence is the public internet. The fence, the internal roads, the gates — that is your VPC infrastructure.



Key Concepts — The Core Building Blocks



Study tip: Visualise each concept as a physical layer of a building. VPC = the building. Subnets = floors. Route Tables = the internal directory. Gateways = the doors.

VPC (Virtual Private Cloud)

Your own isolated, software-defined network inside an AWS Region. You control the IP address space, subnets, routing, and firewall rules. One VPC spans all Availability Zones in a Region. Nothing can enter or leave your VPC unless you explicitly allow it.

CIDR Block

Classless Inter-Domain Routing — the method used to define your VPC's IP address range. Written as an IP address and prefix length: 10.0.0.0/16. The prefix defines how many bits are fixed (the network) versus variable (the hosts). You choose your CIDR when creating the VPC and cannot change it easily afterward.

Subnet

A subdivision of your VPC's IP space, tied to exactly ONE Availability Zone. You deploy resources (EC2, RDS, Lambda ENIs) into subnets. Subnets are either public (has a route to an Internet Gateway) or private (no direct internet route). One VPC can have many subnets.

Route Table

A set of rules that determines where network traffic from a subnet is directed. Every subnet must be associated with exactly one route table. The route table contains entries like: 'traffic to 0.0.0.0/0 (the internet) goes to the Internet Gateway.'

Internet Gateway (IGW)

A horizontally-scaled, redundant AWS-managed gateway that enables communication between your VPC and the public internet. A public subnet is public BECAUSE its route table has a route pointing to the IGW. Without an IGW route, a subnet is private regardless of any other setting.

NAT Gateway

Allows EC2 instances in PRIVATE subnets to initiate OUTBOUND internet connections (e.g., to download software updates) while preventing inbound connections from the internet. The NAT Gateway itself lives in a PUBLIC subnet and has a public IP. Private subnet instances route their internet traffic through it.



Key Concepts — Security Layers & Traffic Control

Security Groups vs Network ACLs — The Most Tested Distinction

Feature	Security Group	Network ACL (NACL)
Level	Instance-level (applied to EC2 ENI)	Subnet-level (applied to entire subnet)
State	STATEFUL — return traffic auto-allowed	STATELESS — must explicitly allow both directions
Rules	Allow rules only — no explicit deny	Allow AND Deny rules
Evaluation	ALL rules evaluated — most permissive wins	Rules evaluated in number order — first match wins
Default inbound	DENY all (nothing allowed until you add rules)	ALLOW all (default NACL allows everything)
Default outbound	ALLOW all outbound	ALLOW all outbound
Use case	Primary layer for instance-level security	Secondary defence — blocking specific IPs/ports at subnet

Additional Networking Concepts:

VPC Peering:

A networking connection between two VPCs enabling them to route traffic to each other using private IPs. Non-transitive — if A peers B and B peers C, A cannot reach C through B.

Elastic IP (EIP):

A static, public IPv4 address that you can attach to an EC2 instance or NAT Gateway. Unlike auto-assigned public IPs, it persists after stop/start. You pay for unattached EIPs.

VPC Endpoints:

Allow private communication between your VPC and AWS services (like S3, DynamoDB) without internet traffic. Gateway endpoints (S3, DynamoDB) are free. Interface endpoints cost per hour.

How It Works — Building a Production VPC Step by Step

1

Design Your IP Space First

Before clicking anything in the console, decide your CIDR block. A /16 VPC (e.g., 10.0.0.0/16) gives you 65,536 IP addresses — more than enough for any production workload. Then plan your subnets: typically a /24 per subnet (256 addresses, 251 usable after AWS reserves 5). Plan at least one public subnet and one private subnet per Availability Zone for high availability. For three AZs: 6 subnets minimum. Draw this on paper first.

2

Create the VPC and Subnets

In the VPC console, click Create VPC. Enter your CIDR block. Name it clearly — e.g., prod-vpc. Then create your subnets: specify which AZ each belongs to and the CIDR range for each (e.g., 10.0.1.0/24 for Public-AZ1, 10.0.2.0/24 for Private-AZ1). Subnets are private by default — do not assume a subnet is public just because it has the word public in its name. Publicness is determined by routing, not naming.

3

Attach an Internet Gateway and Configure Routing

Create an Internet Gateway. Attach it to your VPC. Then create a custom route table for your public subnets — add one route: Destination 0.0.0.0/0, Target: the Internet Gateway. Associate this route table with your public subnets. Your public subnets now have internet access. For private subnets: create a NAT Gateway in a public subnet (requires an Elastic IP). Add a route in the private route table: 0.0.0.0/0 → NAT Gateway. Private instances can now initiate outbound internet traffic but cannot be reached inbound from the internet.



Technical Deep Dive — CIDR Math & Routing Logic

CIDR Notation — How IP Ranges Work

10.0.0.0/16 — Anatomy of a CIDR Block

10.0.0.0 /16 → 65,536 total IPs

10.0.1.0 /24 → 256 IPs (251 usable)

10.0.1.0 /28 → 16 IPs (11 usable)
The Prefix Rule:

- The /N number means N bits are FIXED (network portion)
- Remaining (32-N) bits are variable (host addresses)
- /16 → 32-16 = 16 variable bits → 2^{16} = 65,536 addresses
- /24 → 32-24 = 8 variable bits → 2^8 = 256 addresses
- AWS RESERVES 5 IPs per subnet: .0 (network), .1 (VPC router), .2 (DNS), .3 (future use), .255 (broadcast)
- Smaller prefix = larger range. /8 > /16 > /24 > /28

Public vs Private — It's All About the Route Table

Public Subnet Route Table

10.0.0.0/16	local	All VPC traffic stays internal
0.0.0.0/0	igw-xxxxxx	All other traffic → Internet Gateway

Private Subnet Route Table

10.0.0.0/16	local	All VPC traffic stays internal
0.0.0.0/0	nat-xxxxxx	All other traffic → NAT Gateway (outbound only)

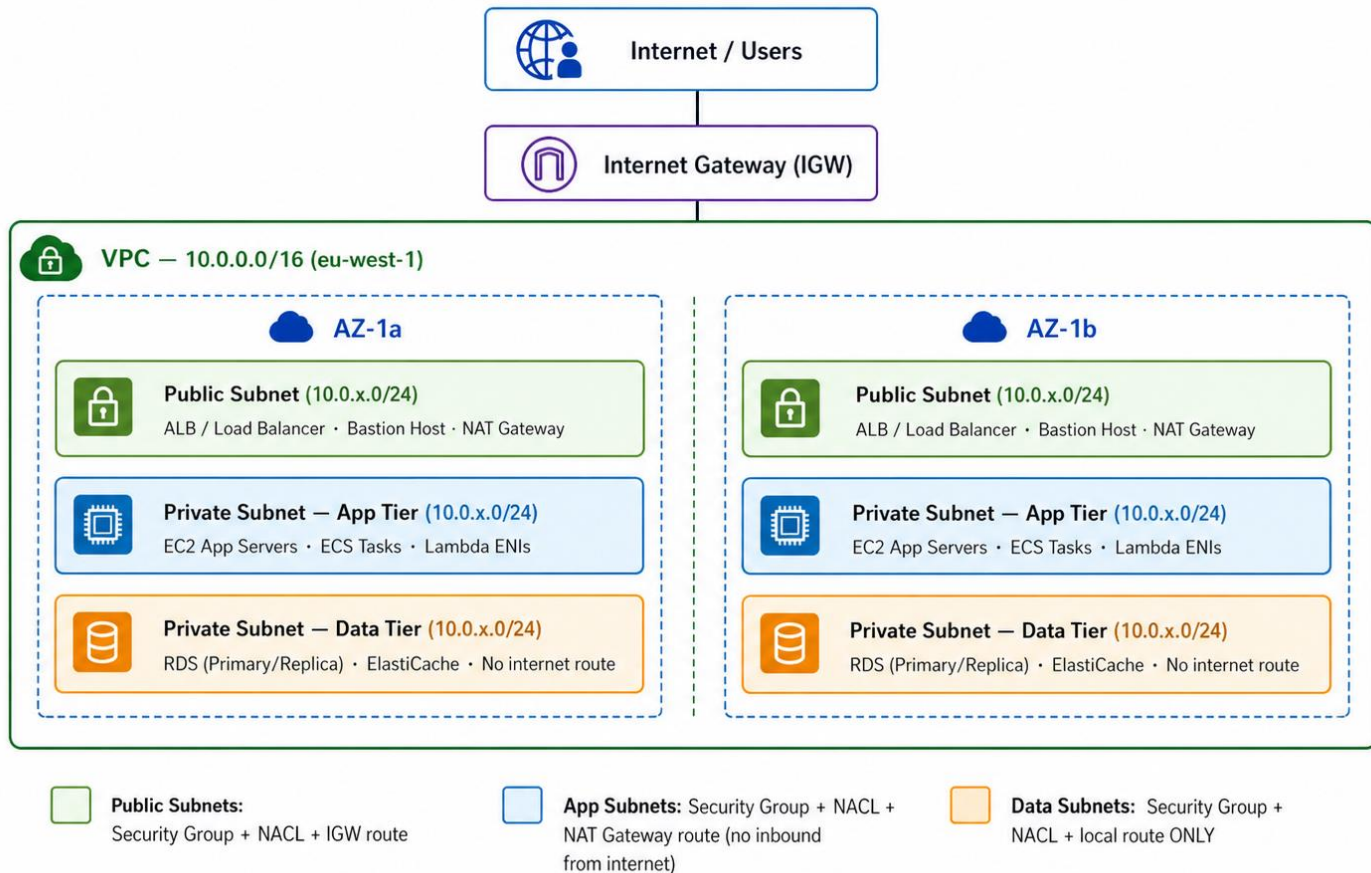
Isolated Subnet Route Table

10.0.0.0/16	local	Only VPC-internal traffic — NO internet route
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Architecture — Production 3-Tier VPC with High Availability

Standard architecture for any production AWS workload: public subnets for internet-facing resources, private subnets for application tier and databases, across 2+ AZs for resilience.





Real-World Nigerian Scenario — Cowrywise-Style Investment Platform

SCENARIO:

A Nigerian retail investment platform — modelled on Cowrywise's architecture — is migrating from shared hosting to AWS. They manage mutual fund investments, fixed deposit products, and savings plans for 500,000 Nigerian retail investors. The SEC (Securities and Exchange Commission) and CBN both require network isolation between customer-facing systems and financial processing backends. A single public data breach would be catastrophic. The platform architect must design a VPC that satisfies both regulatory requirements and high-availability expectations.



Architecture Decisions & Why Each One Was Made

CIDR: 10.0.0.0/16

Choosing 10.0.0.0/16 gives 65,536 IPs — far more than needed today but leaving room for growth without re-architecture. They chose a /16 rather than /24 because Cowrywise expects to add product lines (loans, insurance, pensions) and each new product tier may need dedicated subnets. CIDR blocks cannot be easily changed after VPC creation.

Two public subnets (one per AZ)

Public subnets host only the Application Load Balancer and the Bastion Host. The ALB receives HTTPS traffic from retail investors. The Bastion Host is the ONLY way engineers SSH into the environment — no EC2 instance has a directly assigned public IP. Security Group on the Bastion allows SSH only from the company's office static IP. This limits the public attack surface to two controlled entry points.

Private application subnets

The investment processing API runs on EC2 instances in private subnets. These instances have NO public IPs, no direct internet route. They receive traffic only from the ALB via the ALB's Security Group. Even if an attacker compromises the ALB, they cannot reach the API servers directly without also bypassing the SG rules. The NAT Gateway allows API servers to reach external APIs (like SELAR, Paystack webhooks) without inbound exposure.

Private data subnets — completely isolated

RDS (PostgreSQL) and ElastiCache (Redis session store) live in private data subnets with route tables that have NO 0.0.0.0/0 entry of any kind — not even a NAT Gateway route. The only traffic these subnets handle is within the VPC. The Security Group on RDS allows connections only on port 5432 from the application servers' Security Group. This is the SEC's required network segregation for financial data — provable in architecture diagrams for regulatory audits.



Discussion: Why does putting RDS in a private subnet WITHOUT a NAT Gateway route provide stronger security than one WITH a NAT Gateway route? What attack scenario does removing the NAT Gateway route prevent?



Common Mistakes & Exam Traps

● These are the most common SAA-C03 exam errors on VPC — every one costs marks.

X **TRAP 1: Security Groups are STATEFUL — NACLs are STATELESS. They behave completely differently.**

If you add an inbound allow rule in a Security Group, the return traffic is automatically allowed. You do NOT need a separate outbound rule. NACLs require explicit rules for BOTH directions. If you allow inbound HTTP on port 80 in a NACL, you must also add an outbound rule for the ephemeral port range (1024–65535) or the response traffic is blocked. Candidates who treat NACLs like Security Groups lose marks consistently.

X **TRAP 2: A subnet is public because of its ROUTE TABLE — not because of its name or any other setting.**

You can name a subnet 'Public-Subnet-AZ1' and enable auto-assign public IP — but if its route table has no route to an Internet Gateway, it is a private subnet. Publicness is entirely determined by whether the route table has a 0.0.0.0/0 entry pointing to an IGW. This is the most conceptually important VPC fact. Exam questions will describe a subnet with a public name and test whether you understand that routing determines access.

X **TRAP 3: NAT Gateway provides OUTBOUND internet for PRIVATE subnets — it does NOT allow inbound internet connections.**

A very common distractor: 'Enable a NAT Gateway so users on the internet can access the application servers.' WRONG. NAT Gateway is for instances in private subnets to INITIATE outbound connections (software updates, API calls). Inbound access requires: public subnet + Internet Gateway + Load Balancer or public IP. The NAT Gateway blocks all unsolicited inbound traffic — that's the security point of it.

X **TRAP 4: VPC Peering is NON-TRANSITIVE. You cannot route through a peer to reach another VPC.**

If VPC A peers with VPC B, and VPC B peers with VPC C, instances in VPC A CANNOT reach instances in VPC C through VPC B. Each peering relationship is a direct, private connection between exactly two VPCs. To connect three VPCs to each other, you need three peering connections (A-B, B-C, A-C), or use AWS Transit Gateway. Exam questions describe three-VPC scenarios and ask what's needed — full-mesh peering or Transit Gateway.

X **TRAP 5: VPC spans all AZs in a Region — subnets are scoped to exactly ONE AZ.**

A VPC is a Regional resource. It spans all Availability Zones in that Region automatically. Subnets, however, are zonal — each subnet exists in exactly one AZ. High availability requires creating subnets in at least two AZs and placing redundant resources (EC2 instances, RDS replicas, ALB nodes) across them. Exam questions about 'a VPC that spans multiple AZs' are correct — 'a subnet that spans multiple AZs' is never correct.



Question

A solutions architect at a Lagos fintech is designing an AWS infrastructure for a payment processing application. The EC2 instances running the payment API must be able to download software patches from the internet, but must NOT be directly reachable from the internet. Users must access the application through a load balancer only. Which combination of VPC components correctly meets these requirements? (Select TWO)

A

Place the EC2 instances in a public subnet with a Security Group that blocks all inbound traffic from 0.0.0.0/0.

B

Place the EC2 instances in a private subnet and configure a NAT Gateway in a public subnet. Add a route in the private subnet's route table pointing 0.0.0.0/0 to the NAT Gateway.

C

Place the Application Load Balancer in a public subnet with an Internet Gateway route. Configure the ALB to forward traffic to the EC2 instances in the private subnet.

D

Place the EC2 instances in an isolated subnet with no route table entries and use a VPC Endpoint to allow them to download patches.

E

Enable a VPN Gateway on the VPC and route all outbound traffic from EC2 instances through the corporate VPN to access the internet.



Question

A solutions architect at a Lagos fintech is designing an AWS infrastructure for a payment processing application. The EC2 instances running the payment API must be able to download software patches from the internet, but must NOT be directly reachable from the internet. Users must access the application through a load balancer only. Which combination of VPC components correctly meets these requirements? (Select TWO)

A

Place the EC2 instances in a public subnet with a Security Group that blocks all inbound traffic from 0.0.0.0/0.

Wrong. Public subnet instances have IGW routes. Even if the Security Group blocks inbound, the instances are in a public subnet and could be reached if SG is misconfigured. The requirement is architectural isolation, not just SG rules.

B

Place the EC2 instances in a private subnet and configure a NAT Gateway in a public subnet. Add a route in the private subnet's route table pointing 0.0.0.0/0 to the NAT Gateway.

Correct. Private subnet = no direct inbound from internet. NAT Gateway in public subnet = EC2 instances can initiate outbound connections (patches). Inbound is blocked by design — only the Load Balancer can forward traffic to them.

✓ CORRECT

C

Place the Application Load Balancer in a public subnet with an Internet Gateway route. Configure the ALB to forward traffic to the EC2 instances in the private subnet.

Correct. This is the standard 3-tier architecture. The ALB in the public subnet receives internet traffic via the IGW route. It forwards only to the EC2 target group in the private subnet. Users reach the app through the ALB — never the EC2 directly.

✓ CORRECT

D

Place the EC2 instances in an isolated subnet with no route table entries and use a VPC Endpoint to allow them to download patches.

Wrong. VPC Endpoints connect to AWS services (S3, DynamoDB etc.), not to the general internet. Software patches from package managers (apt, yum) require general internet access. A NAT Gateway is required, not a VPC Endpoint.

E

Enable a VPN Gateway on the VPC and route all outbound traffic from EC2 instances through the corporate VPN to access the internet.

Wrong. A VPN Gateway connects to on-premise networks — it does not provide general internet access for patch downloads. This would route traffic to a corporate network, not to package repositories. Incorrect tool for the requirement.



Key Takeaways — What You Learned Today

VPC Fundamentals

✓ Concepts Covered

- ✓ VPC — your isolated AWS network spanning all AZs in a Region
- ✓ CIDR blocks — /16 gives 65K IPs, /24 gives 256, /28 gives 16
- ✓ Subnets — zonal, public (IGW route) vs private (no IGW route)
- ✓ Route Tables — routing determines public vs private, not naming
- ✓ Internet Gateway — enables two-way internet for public subnets
- ✓ NAT Gateway — outbound-only internet for private subnet instances
- ✓ Security Groups — stateful, instance-level, allow rules only
- ✓ Network ACLs — stateless, subnet-level, allow AND deny rules
- ✓ VPC Peering — non-transitive direct connection between two VPCs
- ✓ Production pattern: ALB in public → EC2 in private app → RDS in private data



Before Next Class

1. Recall all 10 concepts above without notes — active retrieval only
2. Console lab: Create a custom VPC with 2 public and 2 private subnets across 2 AZs
3. Attach an Internet Gateway to your VPC and configure the public subnet route tables
4. Create a NAT Gateway in the public subnet and route private subnet traffic through it
5. Test: launch an EC2 in the private subnet, confirm no inbound internet access but outbound works
6. Re-read the 5 exam traps — write the SG vs NACL distinction 3 times by hand
7. Draw your VPC architecture on paper. Label every component and its IP range.



Further Resources & Reading

Go deeper with these official AWS and industry resources:



Amazon VPC User Guide — Official Documentation

OFFICIAL DOCS

<https://docs.aws.amazon.com/vpc/latest/userguide/>

The authoritative reference for every VPC feature: CIDR allocation, subnet creation, IGW setup, NAT Gateway configuration, security groups, NACLs, peering, and endpoints.

Read the 'Getting Started' and 'Security' chapters before any other resource.



VPC Subnet CIDR Calculator — Visual IP Planner

TOOL

<https://www.subnet-calculator.com/> or cidr.xyz

Practice CIDR math using these interactive calculators. Enter a CIDR block and see exactly how many IPs are available, the network address, broadcast address, and valid host range. Essential for SAA-C03 questions asking how many hosts fit in a /26 vs /28.



AWS VPC Design Best Practices — AWS Architecture Blog

BEST PRACTICES

<https://aws.amazon.com/blogs/networking-and-content-delivery/vpc-subnet-design-best-practices/>

AWS's own opinionated guide to VPC subnet sizing, IP address planning for growth, and the recommended patterns for multi-AZ, multi-tier architectures. The blog post on 'thinking in CIDRs' is particularly strong for exam preparation.



Security Groups vs NACLs — Comparison Guide

EXAM PREP

https://docs.aws.amazon.com/vpc/latest/userguide/VPC_Security.html

The official AWS comparison of Security Groups and NACLs including worked examples showing stateful vs stateless behaviour with ephemeral ports. Understanding the ephemeral port range (1024-65535) for NACL outbound rules is a specific SAA-C03 exam requirement.



SAA-C03 Exam Guide — Domain 1: Resilient Architectures

EXAM PREP

https://d1.awsstatic.com/training-and-certification/docs-sa-assoc/AWS-Certified-Solutions-Architect-Associate_Exam-Guide.pdf

Domain 1 covers multi-AZ VPC design and network resilience — both rely on correct VPC architecture. Download the PDF and read Task Statement 1.1 (design for availability) through 1.3 (design high-performing architectures). VPC appears in almost every domain.



Homework: Read at least ONE resource above before next class. Post what you learned on the WhatsApp group.



Week 3 · Day 1 Complete

VPC Fundamentals

Networking & VPC

Next: Hands-on Console Lab →

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